

Arnav Khandelwal

Software Engineer

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ABOUT

Software engineering undergraduate with hands-on experience building backend applications, REST APIs, database-driven workflows, and containerized services using Python and Java. Strong in debugging, performance optimization, API reliability, and translating business workflows into maintainable software, with additional experience in AI-enabled systems.

EXPERIENCE

Backend Engineering Intern — Dehurdle

Jun 2026 – Aug 2026

Python · FastAPI · Pydantic · Docker · Ollama · Qwen3-4B

Project Edge Node

- Developed and deployed **Project Edge Node**, a fully offline FastAPI service for processing workplace telemetry with strict request validation, deterministic structured responses, and local inference.
- Improved service reliability by implementing Pydantic schema validation, structured error handling, retry/backoff, and Docker Compose-based deployment for reproducible local execution.
- Reduced mean request latency from ~5.3s to ~1.0s (~5.3×) by diagnosing an IPv6-first *localhost* DNS resolution bottleneck and switching service communication to 127.0.0.1.
- Built a 50-request performance benchmark tracking TTFT, end-to-end latency, RAM, and VRAM usage to evaluate runtime configurations and select the final Qwen3-4B Q4_K_M deployment setup.

PERSONAL PROJECTS

Cited Multi-Agent Researcher

Jun 2026

Python · FastAPI · Gemini · asyncio · Pydantic · REST APIs

Agent

- Reduced five-worker execution time from 152s to 26s (5.7x) by diagnosing blocking SDK calls and offloading them to worker threads using `asyncio.to_thread`.
- Developed an asynchronous FastAPI application that decomposes requests and executes up to five independent service workers concurrently using `asyncio.gather`.
- Built deterministic source deduplication, citation assignment, and credibility-scoring logic backed by automated unit tests.
- Improved third-party API resilience through structured response validation, exponential backoff, rate-limit failover, and error handling.

BusinessHQ — Enterprise Operations & Integration Platform

Aug 2026

Java · Spring Boot · Spring Data JPA · MySQL · REST · SOAP · JUnit · Docker

BuisnessHQ

- Built a Spring Boot internal business application to automate vendor onboarding, approval, and operational request workflows using layered Controller–Service–Repository architecture and explicit workflow-state validation.
- Integrated external REST and SOAP services for vendor registration and verification, implementing request validation, timeouts, retry/backoff, and structured failure handling for dependent-service failures.
- Improved workflow-query performance from 180 ms to 35 ms across 10,000 seeded records by profiling MySQL queries and adding targeted indexes through Spring Data JPA.
- Validated 35+ business and integration scenarios using JUnit and Mockito, covering duplicate-vendor detection, approval transitions, external-service failures, and transactional consistency.

TECHNICAL SKILLS

Languages: Python, Java, SQL

Core Computer Science: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks

Application & Backend: Spring Boot, Spring Data JPA, FastAPI, RESTful APIs, Web Services

Databases: SQLite, MySQL, PostgreSQL

Web Services & Integration: REST, SOAP, XML, JSON, HTTP/HTTPS, Third-Party API Integration

Testing & Tools: JUnit, Mockito, pytest, Unit Testing, Integration Testing, API Testing, Git, GitHub, Maven, Postman

AI / ML: PyTorch, Transformers, LLMs, Multi-Agent Systems, RAG, Embeddings, LangChain, Ollama

Systems: Linux, Docker, Docker Compose, TCP/IP, WebSockets, Debugging, Performance Profiling, Retry/Backoff

EDUCATION

Malaviya National Institute of Technology, Jaipur

Aug 2023 – Present

B.Tech in Electrical Engineering

Tagore Public School

Apr 2021 – Mar 2023

CBSE Class XII – PCM

93.4%

ACHIEVEMENTS

Amazon ML Summer School'25

Aug 2025

JEE Main 2023: 98.6 Percentile

2023

Asian Championship Skating: Bronze Medal

Nov 2018